FLOORPLANNING FOR EDA • LAB 07

Simulated Annealing Floorplan Search

Toy cost adds one thousand when a packing is illegal

---

learn_floorplanning — 07 simulated annealing fp

ALGORITHM SPECIFICATION

Pseudocode Workflow

```
INPUT: Modules, Temp T_0, Cooling Rate alpha
cost = deadspace + HPWL + 1000 * (!legal)

WHILE T > T_min DO:
  S_cand = saSwap(S_curr)  // Swap origin coordinates
  delta_C = cost(S_cand) - cost(S_curr)
  IF delta_C <= 0 OR rand() < e^(-delta_C/T):
    S_curr = S_cand
    IF legal(S_curr) AND cost(S_curr) < cost(S_best):
      S_best = S_curr
  T = T * alpha
```

TRY THESE

Algorithm sketch

```
INPUT: pack / representation, T schedule
OUTPUT: best legal low-cost pack
cost ← 1000·¬legal + deadspace + α·HPWL
propose neighbor (swap/move/perturb)
accept if Δ<0 or rand < e^(−Δ/T)
keep best; cool T
GOLDEN legal cost≈36; bad≈1044
```

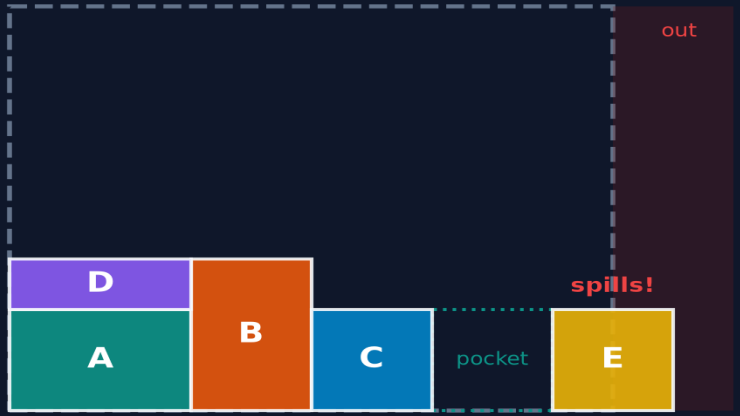
Try: (1) start from the overflow seed and read cost ≈ 1044. (2) saSwap E into the empty pocket — legality true, cost ≈ 36. (3) worsening moves only with e^(–ΔC/T); rescore after every swap; do not quench.

STATE TRANSITION

Bad Seed vs. Golden Packing

Step 1: Bad Overflow Seed

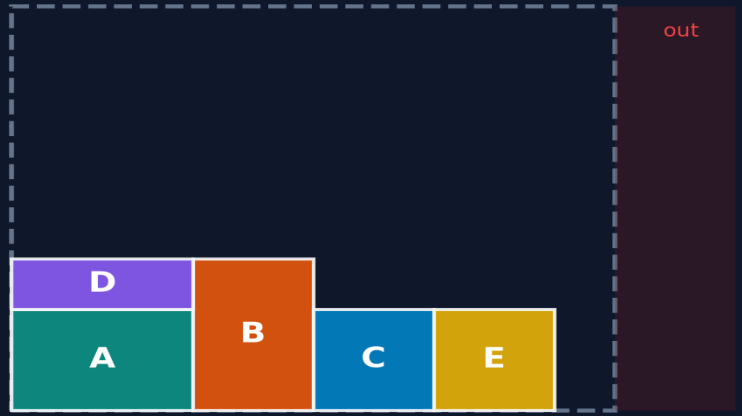
Illegal placement incurs penalty tax



Cost: ~1044 | Legality: False

Step 2: Golden Legal Packing

Zero overlap; cost drops below penalty floor



Cost: ~36 | Legality: True

DEBUGGING & CHECKLIST

Implementation Pitfalls

- **Accepting illegal states without penalty:** Leaves solver stuck in invalid layouts.
- **Forgetting to rescore:** Evaluating Metropolis state with stale costs.
- **Quenching (Cooling too fast):** Traps search in local minima prematurely.
- **Missing the empty-pocket move:** Pairwise swaps cannot fill the hole.